

Article

Time-Dependent Evolution of 1-Pentene Cracking Pathways on H-ZSM-5 Zeolite: Role of Olefin Adsorption and Diffusion

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Abstract

While temperature and acidity dominate the design of zeolite catalysts for olefin cracking, the role of reaction time as an independent variable governing pathway dynamic remains elusive. This study integrates experimental and simulation methods to unravel the dynamic competition among carbenium ion cracking, thermal cracking and Confined Catalytic Radical (CCR) pathways during 1-pentene cracking on H-ZSM-5 zeolite at 650 °C. Analysis of the optimum performance envelope (OPE) curves for cracking products revealed that, in the initial reaction stage, the CCR mechanism significantly enhances ethylene yield. As the reaction time prolongs, C₅+ olefins in the gas phase undergo further cracking on the zeolite surface, markedly increasing the contribution of the carbenium ion pathway. Molecular simulations indicate that C₅+ olefins exhibit stronger adsorption capacity but lower diffusion coefficients on H-ZSM-5, and this adsorption–diffusion disparity is a key factor influencing the evolution of 1-pentene cracking pathways. Concurrently, thermal cracking reactions are also enhanced with increasing residence time, which is unfavorable for ethylene formation. This work elucidates the time-dependent evolution of 1-pentene cracking pathways and the regulatory role of intraparticle mass transfer, providing a theoretical basis for optimizing light olefin selectivity through the adjustment of reaction time and catalyst structure.

Keywords: adsorption; diffusion; high temperature; catalytic cracking; light olefins; confined catalytic radical mechanism; H-ZSM-5 zeolite



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1. Introduction

In recent years, increasingly stringent global environmental regulations have led to a more diversified energy consumption structure [1], resulting in reduced fuel demand and a significant increase in demand for chemical products. Consequently, the importance of producing chemicals from traditional fossil energy sources has become increasingly prominent. Light olefins, represented by ethylene and propylene, serve as crucial basic feedstocks in the petrochemical industry and are widely utilized in the production of various chemicals and intermediates [2,3]. Traditionally, light olefins are primarily produced through naphtha steam cracking, a process that operates at high reaction temperatures (>800 °C), generates substantial CO₂ emissions, and exhibits relatively high overall energy consumption [4,5]. This approach no longer aligns with the requirements of green and sustainable development. In comparison, catalytic cracking can operate at lower temperatures

and offers broader feedstock flexibility. Representative technologies such as Deep Catalytic Cracking (DCC) and Catalytic Pyrolysis Process (CPP) can efficiently convert heavy hydrocarbons or olefin-rich feedstocks within the temperature range of 500–550 °C, significantly enhancing both the cleanliness and economic efficiency of production. Compared with alkanes, olefins as catalytic cracking feedstocks exhibit higher reactivity, enabling higher carbon and hydrogen utilization efficiency while reducing energy consumption, thereby demonstrating broad development prospects [6–8].

To address the existing issues in olefin cracking technology, such as inflexible regulation of light olefin yield and low control efficiency in the precise cleavage of C-C bonds, Xu et al. [9,10] proposed the Targeted Catalytic Olefins (TCO) process. This technology integrates principles from naphtha steam cracking (NSC), catalytic cracking, and olefin cracking. Using C₅–C₈ olefins as feedstocks, it enables selective cracking over mesoporous H-ZSM-5 zeolite within the temperature range of 600–700 °C. By suppressing thermal cracking reactions and reducing methane yield, the process achieves precise cleavage of C-C bonds and enhances carbon atom economy [9,11]. As the technology requires efficient cracking of olefins at high temperatures [2,12,13], it is essential to elucidate the reaction mechanism of olefin cracking under the synergistic effects of high temperature and confined catalysis.

Traditionally, the cracking of olefins has been considered to follow the carbenium ion mechanism, which can be divided into two pathways: monomolecular cracking and bimolecular cracking. In the monomolecular pathway, an olefin first forms a carbenium ion on the Brønsted acid sites of the zeolite, followed by β -scission to generate smaller olefins. In the bimolecular reaction, the olefin is initially protonated to form a carbenium ion, which then undergoes oligomerization with another olefin molecule to produce a higher-carbon-number carbenium ion, followed by subsequent cracking [2]. Monomolecular cracking is the predominant route for the cracking of olefins with five or more carbon atoms. For short-chain olefins such as butene, bimolecular cracking plays a major role. In the case of pentene cracking, at lower reaction temperatures (<600 °C), pentene tends to undergo bimolecular cracking, yielding higher proportions of propylene and butene. As the reaction temperature increases, monomolecular cracking gradually becomes dominant, leading to a significant increase in ethylene selectivity [2,10]. When the temperature is further raised (>670 °C), substantial formation of methyl carbenium ions results in increased methane production, and the rate of increase in ethylene selectivity slows [11]. Recently, Yang et al. [14] reported a novel catalytic cracking mechanism for olefins over H-ZSM-5 zeolite, termed the “Confined Catalytic Radical (CCR)” mechanism. Through a combination of in situ spectroscopic analysis and theoretical simulations, it was demonstrated that at high temperatures (>650 °C), the Brønsted acid sites (BAS) in the zeolite can undergo dehydrogenation to form non-acidic [AlO₄]⁰ sites in situ. Then, the 1-pentene molecules adsorb onto the [AlO₄]⁰ sites, where C-H bonds cleavage occur, producing gaseous pentenyl radicals (C₅H₉•) and concurrently regenerating the BAS. The pentenyl radicals subsequently undergo confined cracking reactions within the zeolite channels. In this pathway, the energy barrier for ethylene formation is lower than that for propylene and methane, enabling a molar ethylene-to-propylene (C₂H₄/C₃H₆) ratio above 1.0 while maintaining low methane selectivity, thereby exceeding the limits predicted by the classical carbenium ion mechanism. These characteristics of 1-pentene cracking make it an ideal model substrate for studying catalytic cracking reaction processes. Indicators such as the propylene/ethylene (P/E) ratio and the ethylene/methane (E/M) ratio can be used to quantitatively characterize the dominant mechanistic features and their evolution [15]. Although significant progress has been made in the study of 1-pentene cracking over H-ZSM-5, current research has primarily focused on the effects of static factors, such as

temperature and catalyst properties, on product distribution. The dynamic evolution of reaction pathways with reaction time remains insufficiently explored. The specific mechanisms underlying the time-dependent changes in cracking product distribution are still not well understood, and the impact of secondary reactions involving intermediate products on the dynamic evolution of 1-pentene cracking pathways particularly requires further clarification.

The interaction between the feedstock or intermediate species and the zeolite governs their adsorption and diffusion behavior, thereby influencing the cracking reactivity and reaction pathways. Due to the instability of olefinic reactants and products at high temperatures, experimental approaches face challenges in determining the impact of their adsorption and diffusion processes on olefin cracking reactions. Molecular simulation techniques offer a means to investigate these effects by analyzing the adsorption and diffusion behavior of reactant and product molecules on the catalyst. Liu et al. [16] employed simulations to study the diffusion behavior of reactant and product molecules involved in carbonylation over mordenite (MOR), revealing that the eight-membered ring channels with lower energy barriers serve as the primary sites for carbonylation, while the slightly larger twelve-membered ring channels function as transport pores for the reacting species. Liu et al. [17] combined density functional theory with MD simulations to elucidate how subtle differences in the channel structures of ZSM-12 and ZSM-22 zeolites influence the dual-cycle reaction process in the methanol-to-olefins (MTO) reaction. Bu et al. [18] demonstrated that hierarchical ZSM-5 zeolites with micro-mesoporous structures significantly enhance the diffusion rates of aromatic molecules such as benzene, naphthalene, and anthracene. Huang et al. [19] employed molecular dynamics simulations to investigate the aldol condensation reaction occurring on hierarchical ZSM-5 zeolites, predicting the effect of differences in the diffusion rates of reactant and product molecules on product selectivity. Existing studies often investigate the adsorption/diffusion and reaction of species within zeolites in isolation; however, the coupled application of these phenomena on zeolites remains to be further explored. Therefore, a deeper investigation into the adsorption behavior and diffusion properties of feedstocks and products on zeolites under TCO conditions, coupled with their correlation to experimental performance, will contribute to an enhanced understanding of the cracking behavior of larger molecules and provide guidance for the development of related catalysts.

This study advances beyond the traditional steady state analysis framework by introducing the time dimension of reaction-pathway competition to address this key scientific question. Employing a transient-kinetics approach centered on Optimum Performance Envelope (OPE) curves analysis, combined with Grand Canonical Monte Carlo (GCMC) and molecular dynamics (MD) simulations, the work reveals the real-time shift of 1-pentene cracking from initial dominance of the CCR pathway toward enhanced bimolecular pathway contribution, establishes the structure-activity relationship between the strong adsorption and slow diffusion of C_5+ intermediates and the observed pathway evolution, and on this basis proposes a conceptual time-diffusion-selectivity framework that links reaction time, diffusion constraints, and product selectivity evolution. While currently qualitative, this framework provides a useful basis for understanding the dynamic competition among cracking pathways. This work not only clarifies the microscopic mechanism by which re-adsorption of intermediates regulates the evolution of cracking pathways, but also establishes reaction time and diffusion length as key tunable parameters for controlling catalytic selectivity, thereby providing a theoretical foundation for precise optimization of reaction processes and catalyst design.

2. Results and Discussion

2.1. Characterization of H-ZSM-5 Zeolite

The XRD pattern and Rietveld refinement results of the H-ZSM-5 zeolite are shown in Figure S1 and Table S1, respectively. The zeolite exhibits characteristic peaks of the typical MFI structure with a preferred orientation along the (001) plane. The goodness of fit indicators confirm the refinement quality [20], and the obtained crystal structural parameters are in agreement with those reported in the literature [21–23]. The FTIR spectrum of the sample is presented in Figure S2, and the corresponding peak assignments are summarized in Table S2, which display typical adsorption bands characteristic of the MFI framework. The N₂ adsorption–desorption isotherm of H-ZSM-5 zeolite (Figure S3c) exhibits combined characteristics of Type I and Type IV isotherms with an H4-type hysteresis loop, indicating the presence of both micropores and mesopores in the zeolite structure [16,20]. The BJH pore size curves of the H-ZSM-5 zeolite (Figure S3a,b) indicate a relatively concentrated pore size distribution, with a most probable pore size of approximately 2 nm. This suggests that the material is predominantly microporous, which is consistent with the pore size data determined by BET (Table 1). A peak near 4 nm was observed in the desorption branch, but not in the adsorption branch, indicating this peak is a phantom peak caused by the tensile strength effect [24]. Therefore, the hysteresis loop observed for H-ZSM-5 in the high-pressure region ($P/P_0 > 0.4$) can be attributed to capillary condensation within its mesoporous structure, which is formed by interparticle voids among the zeolite nanocrystals [20,25]. The SEM images of H-ZSM-5 zeolite is shown in Figure S4, the surface of the zeolite crystal particles is relatively regular, with agglomeration observed.

Table 1. Textural properties of H-ZSM-5 zeolite.

Average Pore Size (nm) ¹	S _{BET} (m ² /g) ¹	S _{micro} (m ² /g) ²	V _{total} (cm ³ /g) ³	V _{micro} (cm ³ /g) ⁴
2.285	299.51	285.56	0.171	0.142

¹ Determined by the BET method. ² Calculated by the t-plot method. ³ Calculated according to the quantity of N₂ adsorbed at $P/P_0 = 0.982$. ⁴ Calculated by the t-plot method.

The acidic properties of the H-ZSM-5 zeolite were investigated by NH₃-TPD and Py-FTIR, with the corresponding characterization results presented in Figure S5. Deconvolution of the NH₃-TPD profile indicates a ratio of strong to weak acid sites of 0.68. The properties of Brønsted acid sites in the sample were determined based on the intensity of the spectral band at 1540 cm^{−1} [26]. The total Brønsted acid sites were measured to be 0.020 mmol/g, of which 0.004 mmol/g were strong Brønsted acid sites. These results collectively demonstrate that the overall acidity of the sample is relatively weak.

The relatively weak acidity of the H-ZSM-5 zeolite may arise from the phosphorus modification process. Its ²⁷Al MAS NMR spectrum (Figure S5c) exhibits a peak at 54 ppm, attributed to tetrahedrally coordinated Al in lattice positions [6], along with broad peaks at 40 ppm and −13 ppm. The former is assigned to tetrahedrally coordinated Al in (SiO)_xAl(OP)_{4−x} species formed by partial substitution of Si with P atoms [27], whereas the latter corresponds to octahedrally coordinated Al interacting with phosphate species [28]. The peak area ratio of (Al₄₀ + Al_{−13})/Al₅₄ is 0.57, indicating that a substantial proportion of Al is coordinated with P, a process that leads to a reduction in the zeolite acid site density [29].

2.2. Temporal Evolution of the 1-Pentene Cracking Pathway

The reaction pathways of 1-pentene at TCO reaction temperatures are illustrated in Figure 1. Among these, Pathway II-3 is less favorable and its contribution to the product

distribution is negligible [30,31]. Therefore, it can be concluded that ethylene in the products is primarily generated via the CCR and monomolecular cracking pathways, butene is mainly supplied through the bimolecular pathway, while propylene is provided by both the bimolecular and monomolecular pathways. The carbenium ions with carbon numbers higher than C₅ formed in the bimolecular cracking pathway can either desorb to yield C₅+ olefins or undergo further cracking to produce smaller olefin molecules [32].

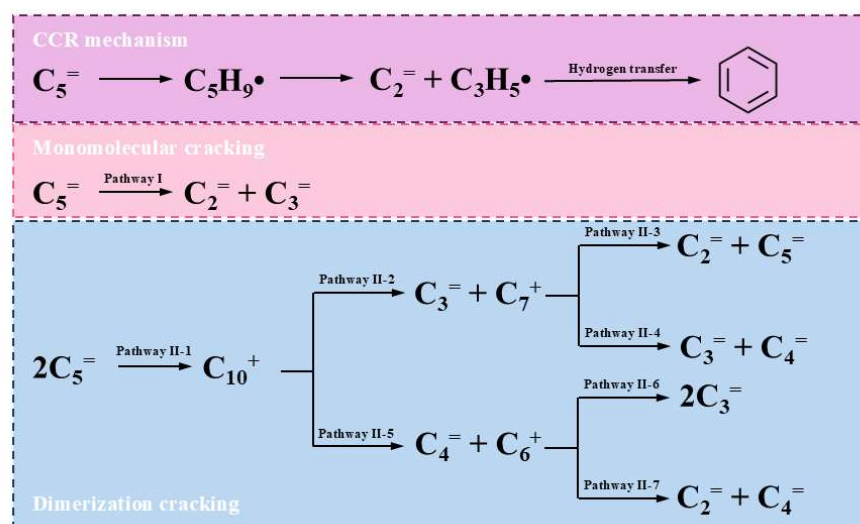


Figure 1. Cracking pathways of 1-Pentene.

At high temperatures, pentene and its cracking products may also undergo free radical thermal cracking and coke formation reactions, yielding products such as methane, 1,3-butadiene, and aromatics. Previous studies [15] have shown that pentene cracking at 650 °C can achieve optimal ethylene selectivity while suppressing methane formation. Therefore, the cracking reactions of 1-pentene at different residence times under this temperature were investigated. The corresponding product composition and selectivity are presented in Figure S6 and Table S3, respectively. In this study, the contact time of 1-pentene on the zeolite was relatively short (space time ranging from 1.77×10^{-2} to 5.31×10^{-2} mg/(mL/min)), under these conditions, the impact of coke deposition or changes in zeolite acidity and surface properties on the reaction can be considered negligible. The primary products from 1-pentene cracking are light olefins (C₂=–C₄=), along with a certain amount of C₅+ olefins. The presence of methane, aromatics, and 1,3-butadiene in the products indicates that the influence of thermal cracking and hydrogen transfer reactions cannot be neglected at this temperature.

The optimum performance envelope (OPE) curve describes the relationship between product yield and feedstock conversion. The slope of its tangent line represents the instantaneous selectivity (S_t) of the product at that conversion. When $X_t \rightarrow 0$, the initial selectivity (S_0) is obtained, which corresponds to the selectivity of the respective product in the initial reaction stage when no gas-phase products are present [33]. Changes in the shape of the OPE curve reflect variations in the product formation pathways over reaction time. Based on the OPE curve, products can be classified as primary or secondary. Primary products are those generated directly from the feedstock on the catalyst surface or formed from intermediates that have not diffused into the gas phase before their desorption. These primary products exhibit a non-zero initial selectivity. Secondary products, however, are formed when gas-phase products re-adsorb onto the catalyst surface, undergo further reactions, and then diffuse back into the gas phase. Their initial selectivity is zero [34,35]. The reaction pathways for primary and secondary products are illustrated in Figure 2.

As conversion increases, deviations from the initial tangent line reveal the contribution of conversion-driven secondary reactions, such as re-adsorption and further cracking of primary products, thermal cracking, or aromatization. Thus, the OPE analysis provides a framework to separate primary products (captured by S_0) from the evolving influence of secondary pathways as residence time and conversion increase. However, it is important to note that this classification is phenomenological and does not by itself constitute proof of a specific reaction mechanism. In this study, the OPE analysis is combined with molecular simulations, stoichiometric correlations, and previous research [6,14,15,32], to arrive at a more robust mechanistic interpretation.

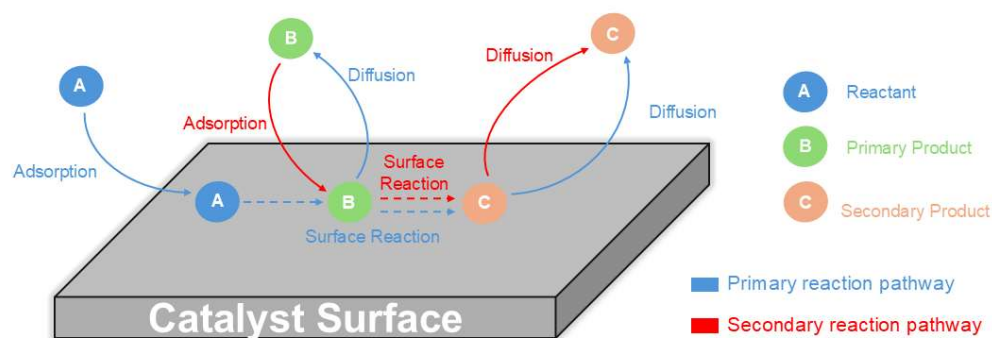


Figure 2. Schematic diagram of primary and secondary reaction pathways in OPE curves.

The OPE curves for cracking products of 1-pentene at 650 °C are shown in Figure 3. The sum of the initial selectivities of all components is approximately equal to 1, indicating that the data distribution is accurate and reliable [36]. Furthermore, the non-zero initial selectivities of all components suggest that each of them includes primary products [34,35], meaning they can be generated directly from 1-pentene on the zeolite surface or via intermediates that have not yet desorbed.

The yield of ethylene maintains a good linear relationship with the conversion of 1-pentene, indicating that it is a stable primary product. The initial mass selectivity ratio of propylene to ethylene is 1.49, meaning that nearly equimolar amounts of ethylene and propylene are generated in the initial reaction stage. However, butene also exhibits a non-zero initial selectivity. In the carbenium ion mechanism, the formation of butene is accompanied by the additional production of propylene (Figure 1, Pathways II-4 to II-6), which would lead to a higher propylene yield than ethylene. Meanwhile, the ratio of the mass selectivity of ethylene to butene is 1.01, i.e., the molar amount of ethylene is twice that of butene in the initial stage, far exceeding the upper limit predicted by the carbenium ion mechanism (Figure 1, Pathway II-7). Therefore, the substantial amount of ethylene observed at the initial reaction stage is likely supplied by the CCR pathway, this interpretation is strongly supported by the recent work of Yang et al. [14], whose DFT and ReaxFF MD simulations show that the CCR pathway has a lower energy barrier for ethylene formation, through which ethylene yield can be significantly higher than that of propylene and butene. The consistency between the product distribution of pentene cracking in this study and CCR characteristics provides robust indirect evidence for its dominance at short reaction times.

The yields of propylene and butene gradually shift above the tangent line of the initial selectivity as the reaction proceeds, displaying characteristics of both stable primary and secondary products [34,35]. This implies that with prolonged reaction time, part of the gas-phase products re-adsorb onto the zeolite surface and undergo further reactions, generating additional propylene and butene.

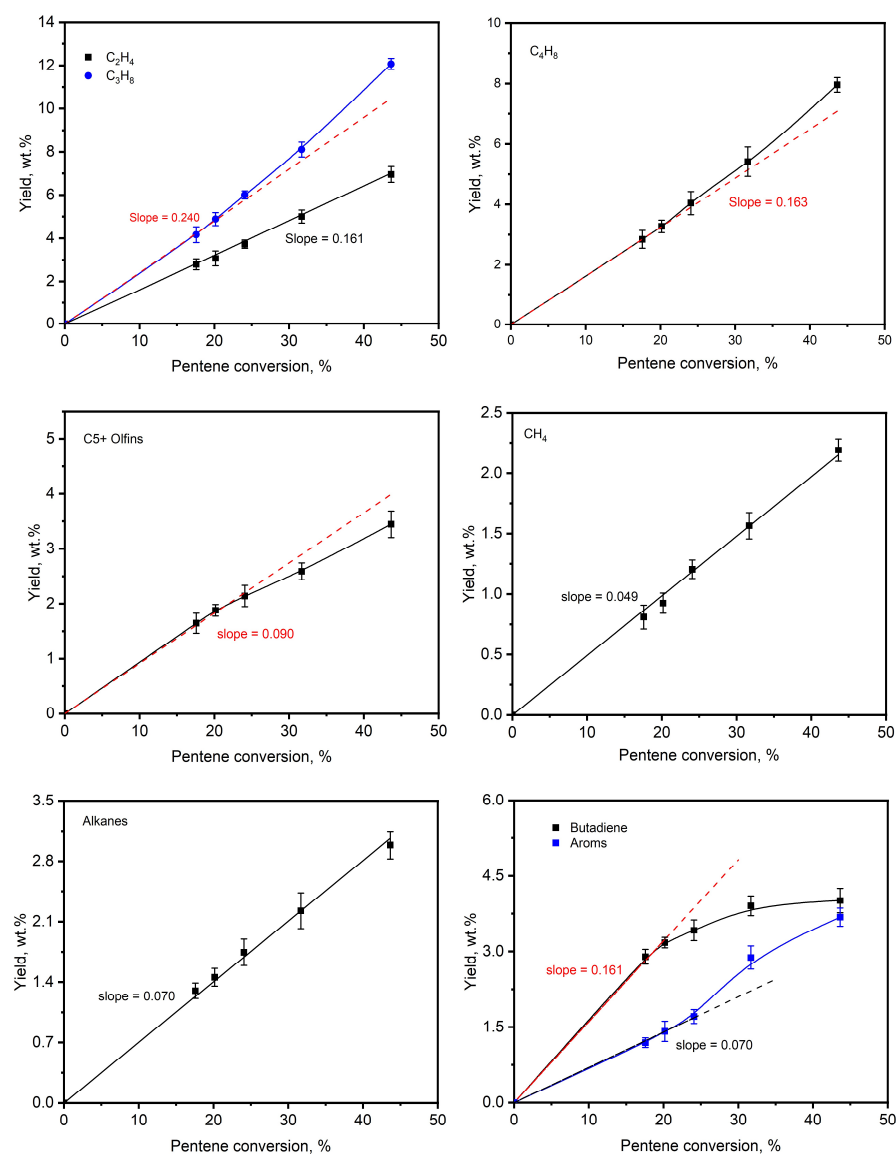


Figure 3. OPE curves of products from 1-Pentene cracking at 650 °C.

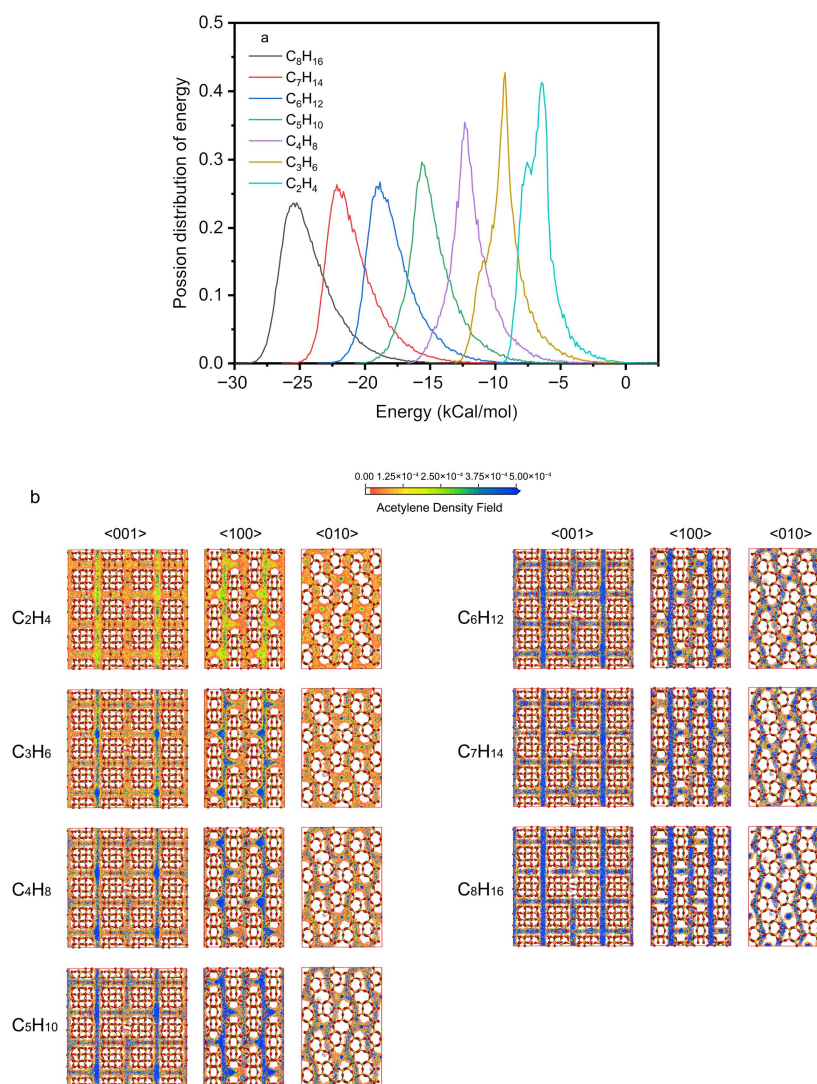
Table 2 uses the DMR index to assess the competitive relationship between the monomolecular/CCR pathway and the bimolecular cracking pathway at different residence times [6], its calculation method is provided in the ESI. It should be noted that the DMR index, originally derived from the stoichiometry of carbenium ion chemistry [6], does not explicitly account for the contribution of the Confined Catalytic Radical (CCR) pathway to ethylene formation. Under the present reaction conditions, the CCR pathway significantly enhances ethylene yield in the initial stage, which may affect the absolute quantitative meaning of the DMR index. Therefore, in this study, the DMR index is employed only as a qualitative indicator to illustrate the increasing contribution of bimolecular cracking with prolonged residence time, consistent with the observed rise in the propylene-to-ethylene (P/E) ratio. This indicates an enhancement of the bimolecular cracking pathway.

While the yields of propylene and butene increase, the OPE curve of C₅+ olefins gradually falls below the tangent line of the initial selectivity, displaying characteristics of an unstable primary product [34,35]. These C₅+ olefins are products of incomplete reactions via the bimolecular cracking pathway of 1-pentene. The change in the OPE curve suggests that some C₅+ olefins in the gas phase may re-adsorb onto the catalyst for further cracking, leading to a more complete bimolecular cracking process.

Table 2. P/E ratio and DMR index for the cracking reaction of 1-Pentene over H-ZSM-5 zeolite at 650 °C.

Temperature, °C	Space Time, $\times 10^{-2}$ mg/(mL/min)				
	1.77	2.13	2.66	3.54	5.31
P/E	1.57	1.59	1.60	1.62	1.65
DMR	0.62	0.63	0.64	0.66	0.69

Accordingly, the interactions between the key intermediate and final olefin products (C_2 – C_8 olefins) from the 1-pentene cracking pathway and the H-ZSM-5 zeolite were investigated by means of molecular dynamics (MD) simulations. As shown in Figure 4a, the interaction energy between olefins and H-ZSM-5 decreases with increasing carbon number. This trend is governed by two main factors controlling olefin adsorption: the van der Waals interactions arising from the mesoporous structure of the zeolite [37], and the covalent or hydrogen bonding interactions between the olefins and Brønsted acid sites [38,39]. When the olefin molecular diameter approaches the pore size of the adsorbent, the molecule can access adsorption sites more easily, resulting in a lower interaction energy with the zeolite and more stable adsorption [40].

**Figure 4.** Interaction energy distribution (a) and adsorption probability density field distribution (at 100 kPa) for C_2 – C_8 olefins on H-ZSM-5 zeolite at 650 °C (b).

The probability density field distributions for single-component olefin adsorption (Figure 4b) reveal that as the carbon number increases, the adsorption probability density of olefins within the zeolite rises, indicating their ability to interact with a greater number of sites inside the zeolite. Table 3 compares the differences between single-component adsorption of C₂–C₈ olefins and competitive adsorption under equimolar conditions, which more closely resemble experimental scenarios. Compared to single-component adsorption, the adsorption capacity of each olefin decreases under multicomponent conditions due to competitive adsorption, with the extent of decrease diminishing as the olefin carbon number increases. This trend occurs because the interaction between olefins and the zeolite strengthens with increasing carbon number, leading to higher adsorption selectivity. The distribution of olefins within the H-ZSM-5 channels is shown in Figure S7, where C₆ and higher olefins occupy the primary adsorption sites in the channels.

Table 3. Adsorption parameters for pure-component C₂–C₈ olefins at 650 °C.

	C ₂ H ₄	C ₃ H ₆	C ₄ H ₈	C ₅ H ₁₀	C ₆ H ₁₂	C ₇ H ₁₄	C ₈ H ₁₆
K_m^1 (Å)	3.75	4.29	4.72	5.09	5.41	5.69	5.95
Q_{ms}^2 (mmol/g)	3.47	1.17	0.69	0.64	0.65	0.55	0.50
Q_{mm}^3 (mmol/g)	0.019	0.020	0.024	0.046	0.067	0.132	0.321
Decline rate (%)	99.45	98.29	96.52	92.81	89.69	76.00	35.80

¹ The kinetic diameter of the olefin molecule (K_m , Å) is estimated using the following equation [41]: $K_m = 1.234\sqrt[3]{M}$ where M is the relative molecular mass of the olefin. ² Maximum adsorption capacity for single-component adsorption ³ Maximum adsorption capacity for equimolar mixture adsorption.

The aforementioned studies elucidate that C₅+ olefins exhibit stronger physical adsorption compared to C₂–C₄ olefins, combined with the OPE curves in Figure 3, these trends suggest that C₅+ olefins may, by virtue of their enhanced physical adsorption, gain greater access to active sites and subsequently undergo further reaction via acid-catalyzed pathways. In contrast, the C₂–C₄ olefins produced from 1-pentene cracking exhibit weaker interactions with the zeolite surface, and their re-adsorption is hindered by the preferential adsorption of higher carbon-number olefins, which suppresses their consumption through secondary reactions.

Based on the above analysis, cracking experiments of C₆–C₈ olefins over H-ZSM-5 zeolite were conducted to verify their contribution to the yields of propylene and butene. The results, shown in Figure 5, indicate that the selectivities for propylene and butene are significantly higher than that for ethylene. This is because the formation of ethylene during the cracking of C₆–C₈ olefins involves the generation of primary carbenium ions, which is energetically less favorable. Thus, in the 1-pentene cracking reaction, the further cracking of C₅+ olefins serves as the primary source of the secondary products, propylene and butene, while simultaneously enhancing the contribution of the bimolecular cracking pathway for 1-pentene.

The OPE curve of methane exhibits characteristics of a stable primary product [34,35], indicating that it is generated from 1-pentene on the catalyst surface. The involved reaction is likely the monomolecular radical thermal cracking of 1-pentene at high temperatures [15], which produces equimolar amounts of methane and 1,3-butadiene. Yang et al. [14] calculated the contribution of each reaction pathway to the product distribution during 1-pentene cracking at high temperatures and found that methane is primarily generated via thermal cracking. However, on H-ZSM-5 zeolite, methane may also originate from the side-chain

cleavage of aromatics generated via hydrogen transfer reactions during coke formation [13], therefore, the contribution of these pathways requires careful analysis.

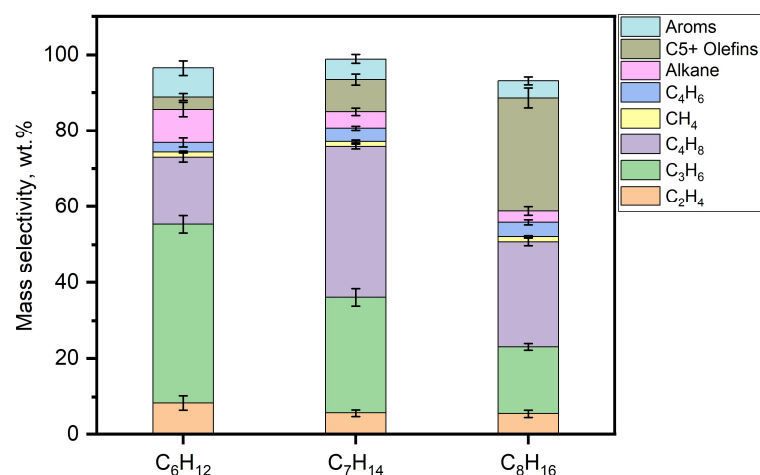


Figure 5. Product mass selectivity for the cracking of C₆–C₈ olefins over H-ZSM-5 zeolite. Reaction conditions: temperature 650 °C; catalyst mass 0.038 g; flow rate 214.51 mL/min; pressure 0.1 MPa.

As indicated by the OPE curves, the initial mass selectivity ratio of 1,3-butadiene to methane is approximately 3.29, which is remarkably close to the theoretical value of 3.37 derived from the stoichiometry of monomolecular radical thermal cracking of 1-pentene, providing strong evidence for the radical cracking pathway as the primary source of methane in the initial stage of the reaction. Furthermore, the coke yield measured throughout the experiments remained consistently below 0.4%, indicating that coke-related pathways contribute negligibly to methane formation. Therefore, the methane in the products originates primarily from the monomolecular radical thermal cracking of 1-pentene.

Since both the CCR and monomolecular carbenium ion cracking reactions that produce ethylene and the free-radical thermal cracking reaction that generates methane are based on a single 1-pentene molecule, the ethylene-to-methane mass selectivity ratio (E/M) can be used to evaluate the competition between catalytic cracking and free-radical thermal cracking. The variation of the E/M value over reaction time for 1-pentene cracking is shown in Figure 6. As the reaction time increases, the E/M value shows a decreasing trend, indicating an enhancement of the pentene thermal cracking reaction, which is unfavorable for ethylene formation.

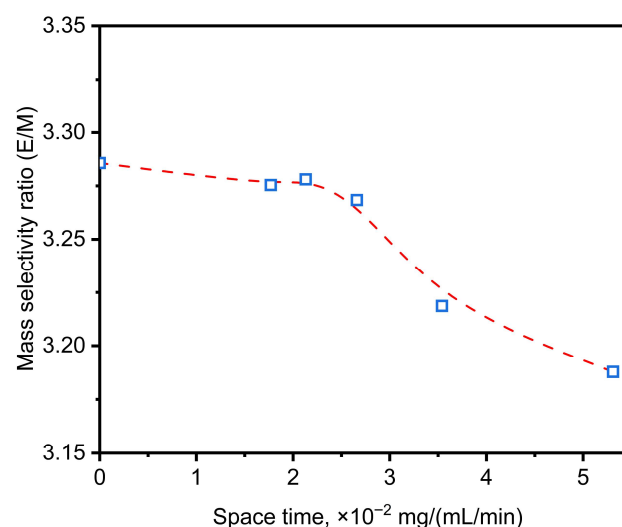
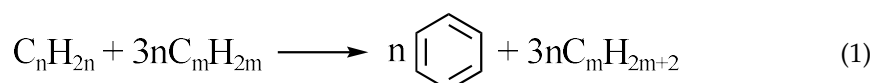


Figure 6. Variation of the E/M over time for 1-Pentene cracking products.

The OPE curve of aromatics in the products exhibits characteristics of a primary product at low pentene conversion. As the reaction time increases, the curve gradually shifts above the tangent line of the initial selectivity, indicating that aromatics are also generated as secondary products. The aromatics categorized as primary products mainly originate from by-products of the CCR pathway (Figure 1), as well as from hydrogen transfer reactions involving either 1-pentene directly on the zeolite surface and/or olefins derived from 1-pentene that remain adsorbed on the H-ZSM-5 surface (Equation (1)). Correspondingly, the OPE curve of the alkanes in the products also exhibits the characteristic features of a stable primary product (Figure 3).



1,3-Butadiene is also an important precursor for aromatics under catalytic cracking conditions [42], and it may contribute to the formation of aromatics through cyclization reactions. As the reaction time increases, the mass yield of 1,3-butadiene shifts significantly below the tangent line of the initial selectivity, indicating that a portion of it is adsorbed onto the H-ZSM-5 zeolite and undergoes further reactions. By combining the OPE curves of aromatics and 1,3-butadiene, it can be inferred that after 1-pentene undergoes thermal cracking to form 1,3-butadiene, the gaseous 1,3-butadiene re-adsorbs onto the zeolite surface with prolonged reaction time, where it reacts further to generate aromatics as secondary products.

2.3. Diffusion Properties of Olefins in H-ZSM-5

As discussed earlier, the re-adsorption of C_5+ olefins during the cracking of 1-pentene is a key factor enhancing the bimolecular pathway. Their strong adsorption affinity increases the probability of being captured by active sites within the zeolite. However, their accessibility to these active sites is determined by their diffusion properties within H-ZSM-5. Through cracking experiments of linear C_5 – C_8 olefins over H-ZSM-5 zeolite, the relationship between the Thiele modulus (φ) and the reaction effectiveness factor (η) was calculated to verify the influence of diffusion of C_5 and higher olefins on the cracking reaction. The results are shown in Figure 7, and the calculation methods for φ and η are provided in the ESI.

When the Thiele modulus is less than 0.2 and the effectiveness factor exceeds 0.99, it indicates that the catalytic cracking reaction operates under a kinetically controlled regime. Conversely, when the Thiele modulus exceeds 5 and the effectiveness factor falls below 0.2, the reaction is under a diffusion-controlled regime. Thiele modulus values lying between these two ranges signify that the reaction is subject to transitional constraints (i.e., between kinetic control and internal diffusion control) [43]. The effectiveness factors for C_5 – C_8 olefins all fall within the transitional regime and decrease with increasing carbon number. This indicates that the diffusion resistance of olefins within the H-ZSM-5 zeolite increases with carbon number.

The self-diffusion coefficients for single-component and multi-component C_2 – C_8 olefins at 650 °C, calculated via MD, are presented in Table 4. Their values, ranging from 10^{-12} to 10^{-10} m^2/s , are consistent with literature reports [44]. The diffusivity of olefins with more than five carbon atoms is significantly lower than that of C_2 – C_4 olefins. Under multi-component diffusion conditions, the self-diffusion coefficients are further reduced because the presence of other components occupies the zeolite channels [44,45]. Consequently, under the experimental conditions for 1-pentene cracking, the C_5+ olefins in the gas phase have difficulty entering the zeolite pores and accessing the active sites due to

their limited diffusivity. This restricts their further conversion, leading to a relatively high yield of C₅+ olefins in the products. Furthermore, the C₅+ olefins generated in the initial reaction stage struggle to diffuse into the gas phase. This not only hinders the diffusion of products such as C₂–C₄ olefins but also increases the probability of side reactions, which is reflected in the non-zero initial selectivities for aromatics and alkanes, as some intermediates are trapped and undergo hydrogen-transfer reactions before diffusing out.

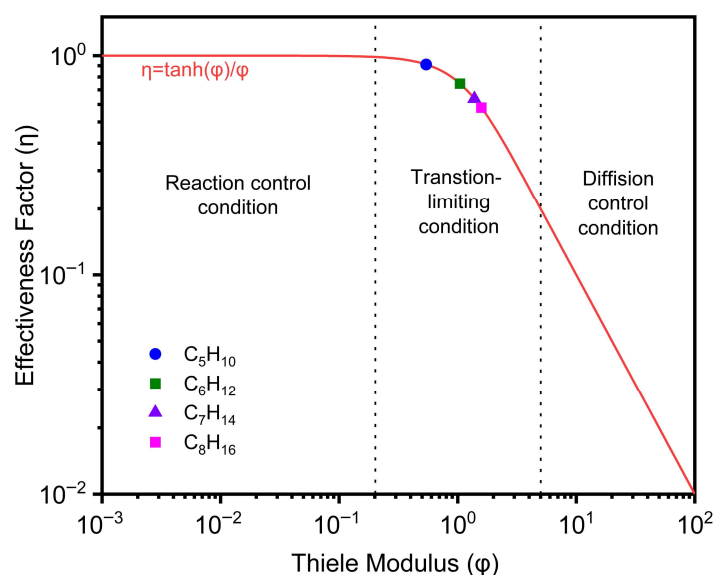


Figure 7. Relationship between the effectiveness factor and the Thiele modulus for the cracking of C₅–C₈ olefins over H-ZSM-5 zeolite.

Table 4. Self-Diffusion Coefficients of Single-Component and Multi-Component C₂–C₈ Olefins on H-ZSM-5 Zeolite at 650 °C.

Olefins	Diffusion Coefficient ($\times 10^{-12}$ m ² /s)	
	Single-Component	Multi-Component
C ₂ H ₄	19.967	2.979
C ₃ H ₆	11.038	2.648
C ₄ H ₈	4.865	2.658
C ₅ H ₁₀	3.396	2.457
C ₆ H ₁₂	2.495	2.437
C ₇ H ₁₄	2.471	1.060
C ₈ H ₁₆	2.423	1.498

The sensitivity analysis results presented in Table S5 indicate that the diffusion of C₅+ olefins in H-ZSM-5 zeolite is less influenced by their cracking reaction rate constant (*k*) and could be instead governed primarily by particle size and pore dimensions. By setting the Thiele modulus for the cracking reactions of C₆–C₈ olefins to 0.2, the required zeolite particle size and pore dimensions necessary to bring these olefins into the kinetically controlled regime were back-calculated [43], the results are presented in Table S4. The table shows that reducing the zeolite particle size or introducing mesopores to increase the average pore size can alleviate diffusion limitations for C₅ and higher olefins, thereby facilitating more complete reactions.

In summary, by integrating the characteristics of product OPE curves and molecular simulation results, the evolution of the 1-pentene cracking process over reaction time can be elucidated. In the initial stage of 1-pentene cracking, the CCR pathway provides a substantial portion of the ethylene. As the reaction time prolongs, the contribution of the carbenium ion cracking pathway increases. This is attributed to the adsorption of C₅+

olefin intermediates from the gas phase onto the H-ZSM-5 surface, where they undergo further cracking, thereby reinforcing the carbenium ion pathway. Concurrently, the thermal cracking of 1-pentene is also enhanced, which competes with the catalytic cracking pathways (CCR and carbenium ion cracking) and suppresses ethylene formation. The aromatics in the initial reaction stage are mainly supplied by CCR reactions and hydrogen transfer. As the reaction proceeds, the aromatization of 1,3-butadiene contributes additional aromatic products. Therefore, appropriately reducing the reaction time can effectively suppress side reactions such as aromatization and thermal cracking. However, it should be noted that, while this study provides mechanistic insights into the time-dependent evolution of 1-pentene cracking pathways, it relies primarily on product distribution analysis, OPE curves, and simulation calculations. Moreover, during prolonged continuous operation, coke accumulation and potential structural changes in the zeolite framework, such as dealumination or the formation of coke, may become considerably significant. Future studies will employ H-ZSM-5 zeolites with varying acid strengths and acid site densities, in conjunction with radical scavenger experiments, in situ Diffuse Reflectance Infrared Fourier Transform (DRIFT) spectroscopy, in situ electron paramagnetic resonance (EPR) spectroscopy, and isotopic labeling techniques. These approaches aim to accurately characterize the transformation between BAS and $[\text{AlO}_4]^-$ sites on H-ZSM-5 zeolite under high-temperature conditions, elucidate the role of zeolite acidity in the competition among reaction pathways, and provide definitive experimental validation for quantitatively distinguishing each reaction pathway. Building on this, the reaction time range will be extended to systematically investigate the impact of catalyst aging on the time-dependent evolution of 1-pentene cracking pathways.

3. Materials and Methods

3.1. Characterization

The phosphorus-modified H-ZSM-5 molecular sieve with a Si/Al atomic ratio of 76 and the 1-pentene raw material (purity 99%) used in this experiment were both provided by Sinopec Petroleum Processing Research Institute Co., LTD. (RIPP, Beijing, China). The X-ray diffraction (XRD) patterns were measured with an EMPYEEAN diffractometer (PANalytical, Almelo, The Netherlands) with a Cu K α radiation ($\lambda = 0.15406$ nm), scanning range of 5 to 140°, scanning rate of 1°/min, and step size of 0.014°. The background of the XRD patterns and unit cell parameters (P21/n space group) were refined by Rietveld refinement using GSAS-II v1.0.1 software [21].

The Fourier Transform Infrared Spectroscopy (FTIR) was determined on the FTIR-8400 infrared spectrometer (Shimadzu, Nakagyo-ku, Japan). The samples were prepared by the KBr pellet method.

The N₂ adsorption–desorption was determined on the ASAP 2420 fully automatic surface area and pore size analyzer (Micromeritics, Norcross, GA, USA). The adsorption–desorption curves of the sample were measured at −196 °C. The specific surface area and the pore structure data of the sample were calculated, respectively, using the Brunauer–Emmett–Teller (BET) and Barrett–Joyner–Halenda (BJH) methods.

The scanning electron microscope (SEM) was observed using the HITACHI S-4800 (ThermoFisher, Waltham, MA, USA) scanning electron microscope with an acceleration voltage of 20.0 kV.

The NH₃ program temperature desorption (NH₃-TPD) was determined on the Autochem II 2920 chemical adsorption instrument (Micromeritics, Norcross, GA, USA). An amount of 0.05 g of the sample was taken and purged with a He gas flow (25 mL/min) at 400 °C for 60 min before cooling to 150 °C, and then adsorbed in a 40 mL/min 10 v% NH₃/He mixed gas for 60 min. Subsequently, a He gas flow was introduced to blow until

the baseline was stable, and then the sample was heated at a rate of 10 °C/min to 600 °C. The signal was recorded by TCD.

The pyridine adsorption infrared spectrum (Py-IR) was measured using the NI-COLET 6700 spectrometer (ThermoFisher, Waltham, MA, USA). The pressed sample was desorbed in the reaction cell at 400 °C and 10^{-6} Pa for 60 min, then cooled to room temperature to record the background data from 1300 to 1700 cm^{-1} . Then, 2.67 Pa of pyridine was introduced, and the system was equilibrated for 30 min before being equilibrated at 200 °C and 10^{-6} Pa for another 20 min. Then, the low temperature spectrum was recorded. Subsequently, the sample was cooled to room temperature and then heated to 350 °C to obtain the high temperature spectrum. The Brønsted (B) and Lewis (L) acid site concentrations of the H-ZSM-5 zeolite were quantified following the method reported by Emeis [46], using molar extinction coefficients of 1.67 $\text{cm}/\mu\text{mol}$ for pyridine adsorbed on B acid sites (1545 cm^{-1}) and 2.22 $\text{cm}/\mu\text{mol}$ for those on L acid sites (1455 cm^{-1}).

Solid-state ^{27}Al NMR spectra were recorded on a Bruker AVANCE III 500 WB spectrometer (Billerica, MA, USA) under magic angle spinning (MAS) at the resonance frequency 156.4 MHz with a spinning rate 13 kHz.

3.2. Computational Details

The H-ZSM-5 model was constructed based on the purely siliceous MFI unit cell provided by the Materials Studio 2023 (MS) software. Its framework consists of straight channels (5.3×5.6 Å) and sinusoidal channels (5.1×5.5 Å), both formed by ten-membered rings. To ensure the accuracy of the calculations, the dimensions of the simulation cell along all three coordinate axes should be greater than twice the cutoff radius. Therefore, the original unit cell model was expanded into a $2 \times 2 \times 2$ supercell model [45,47]. According to Löwenstein rule [48], Al atoms were randomly substituted for part of the Si atoms while avoiding the formation of Al-O-Al linkages. The resulting negative charge was balanced by protons bonded to the adjacent oxygen atoms, thereby constructing an H-ZSM-5 zeolite model with a Si/Al atomic ratio of 76, as shown in Figure 8.

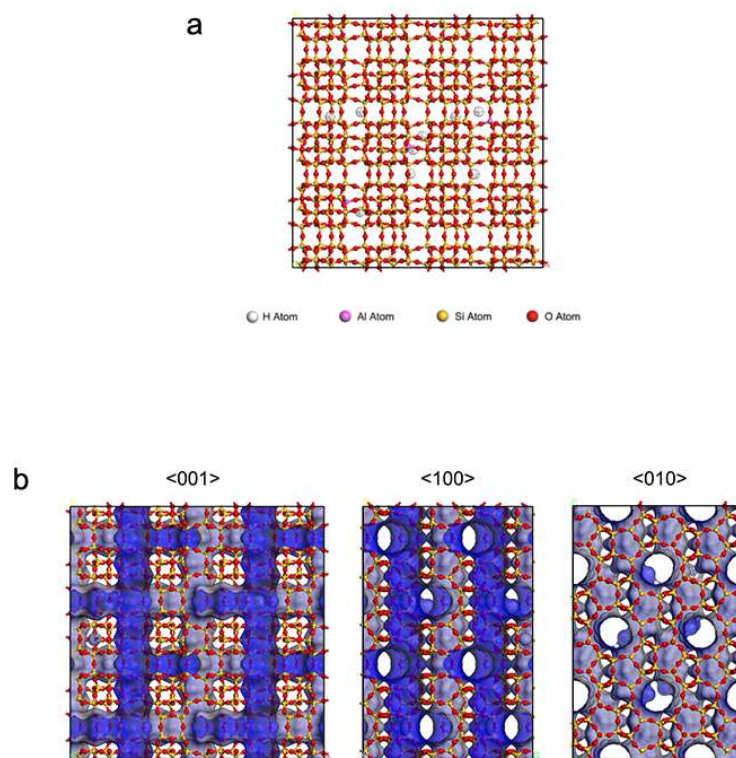


Figure 8. The supercell structure model (a) and pore structure (b) of H-ZSM-5 zeolite.

To validate the accuracy of the model, the XRD pattern of the model was simulated using the Reflex module in Materials Studio [44,45,47]. The simulated pattern was compared with the standard XRD pattern from the International Zeolite Association (IZA) database [49] and the experimental XRD pattern of the actual sample. As shown in Figure S1a, the peak positions and intensities of the simulated pattern align well with both the experimental and IZA standard patterns, exhibiting characteristic peaks typical of an MFI-structured zeolite (PDF: 44-0003). This confirms that the constructed model correctly represents the structure of H-ZSM-5.

Based on the Sorption module in Materials Studio [44,50,51], the physical adsorption process of reactant and product molecules (C_2 – C_8 olefins) on the zeolite was calculated using the Grand Canonical Monte Carlo (GCMC) method. This ensemble operates under the following assumptions: (1) the chemical potential (μ), temperature (T), and volume (V) are fixed; (2) the zeolite framework is treated as rigid within the adsorption system, with changes only occurring in the number of adsorbate molecules; (3) each simulation cycle randomly executes one of four types of moves: translation, rotation, insertion into the framework, or deletion from the framework. The interaction energy (U_{ij}) between adsorbates and the zeolite framework includes contributions from electro-static and van der Waals potentials, described by the Coulomb potential and the Lennard-Jones (L-J) potential, respectively:

$$U_{ij} = 4\epsilon_{ij} \left[\left(\frac{\delta_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\delta_{ij}}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} \quad (2)$$

where ϵ_{ij} and δ_{ij} are L-J parameters, r_{ij} is the distance between atoms i and j , q_i and q_j are the charges of atoms i and j , respectively, and ϵ_0 is the permittivity of vacuum.

The COMPASS III force field was employed to calculate the aforementioned parameters [45,52,53]. The electrostatic interactions were calculated using the Ewald summation method with an accuracy of 1×10^{-5} kcal/mol, while the van der Waals interactions were calculated using the atom-based summation method with a cutoff radius of 12.5 Å.

Based on the Forcite module in MS [51], molecular dynamics (MD) simulations were used to study the diffusion of reactant and product molecules within the zeolite. The NVT ensemble was applied to maintain a constant number of particles, volume, and temperature [45], with temperature controlled using the Nosé method [45,54]. The interactions between the zeolite and the reactant/product molecules were described using the COMPASS III force field [45,53]. To investigate single-component diffusion, 15 molecules were first inserted into the supercell model. The initial configuration for diffusion was then obtained through geometric optimization and annealing using energy minimization, followed by MD calculations. For multi-component diffusion, five molecules of each type were inserted into the supercell model, with subsequent steps being consistent with those for single-component diffusion [45]. Initially, a 100 ps simulation was performed to equilibrate the system, followed by a 1 ns production run to compute the diffusion process. The integration time step was set to 0.1 fs, and atomic trajectories were saved every 2500 steps [45]. The mean squared displacement (MSD), which describes the distance between a molecule's position at time t and its initial position at t_0 , was used in conjunction with the Einstein equation to obtain the self-diffusion coefficient (D_s) [45,53]:

$$D_s = \frac{1}{2dN} \lim_{t \rightarrow \infty} \frac{d}{dx} \sum_{i=1}^N \left\{ \left[\vec{r}_i(t) - \vec{r}_i(0) \right]^2 \right\} \quad (3)$$

where d is the system dimension, N is the number of particles, $[\vec{r}_i(t) - \vec{r}_i(0)]^2$ represents the MSD, taking $1/6$ of the slope of the MSD- t curve is D_s .

The simulations were designed to investigate trends in physical adsorption and diffusion arising from pore confinement and molecular size, thereby providing a basis for interpreting the experimental observations. Due to the nonpolar of olefins, these physical adsorption and diffusion processes are less influenced by the acidic properties of the zeolite [55–57].

3.3. Reaction Set-Up and Conditions

The catalytic cracking of 1-pentene was investigated using a micro fixed-bed reactor system (CDS6200, Oxford, PA, USA) coupled with an online gas chromatograph (Agilent 8890, Santa Clara, CA, USA) equipped with both a flame ionization detector (FID) and a thermal conductivity detector (TCD). A schematic of the experimental setup is shown in Figure 9. The system comprises three main sections: a preheating zone, an isothermal reaction zone, and an online product analysis zone.

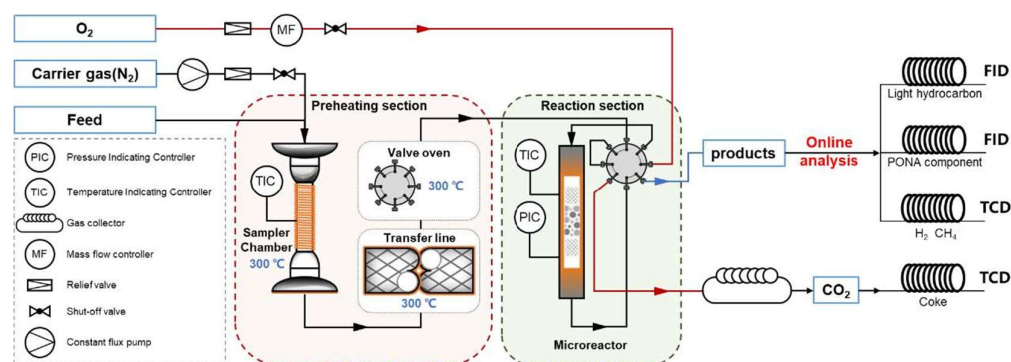


Figure 9. Flowchart of the experimental setup.

The cracking reactions were carried out in a quartz tubular (4 mm inner diameter, 153 mm length). The average particle size of the H-ZSM-5 zeolite is approximately 38.54 μm , so that the ratio of the reaction tube length, diameter to the average particle size of the catalyst is greater than 50 and 10, respectively, thereby satisfying the criteria for plug-flow hydrodynamics and minimizing axial dispersion [58]. The reactant, 1-pentene (99% purity), was fed into the preheater via a calibrated micro-syringe pump, vaporized, and then directed through the catalyst bed maintained at the target temperature. The effluent stream was analyzed in real-time by the online GC. After each reaction run, catalysts were regenerated by coke combustion at 700 $^{\circ}\text{C}$ under O_2 flow. CO_2 from combustion was quantified using a universal gas collector to determine coke yield. This configuration enabled complete product speciation and quantification.

Due to the short residence time, the hydrogen and coke yields were lower than 0.4 wt%, so their effects on the cracking reaction were ignored [15]. On ZSM-5 zeolite, the apparent activation energy for pentene isomerization is typically 20–40 $\text{kJ}\cdot\text{mol}^{-1}$ lower than that for its cracking reaction, while the reaction rate constant is one to two orders of magnitude higher. This ensures rapid equilibration among various pentene isomers [15]. Thus, conversion is defined as the reaction process wherein 1-pentene produces all products other than its isomers.

The conversion of 1-Pentene X_{Pentene} , the mass yield $Y_{(i)}$, the mass selectivity $S_{(i)}$, and the molar selectivity $S_{M(i)}$ of each product were calculated by the following equations.

$$X_{\text{Pentene}} = \frac{W_{\text{Pentene, in}} - W_{\text{Pentene, out}}}{W_{\text{Pentene, in}}} \times 100\% \quad (4)$$

$$Y_{(i)} = \frac{W_{(i)}}{W_{\text{Pentene, in}}} \times 100\% \quad (5)$$

$$S_{(i)} = \frac{Y_{(i)}}{X_{\text{Pentene}}} \quad (6)$$

$$S_{M(i)} = \frac{n_{(i)}}{n_{\text{Pentene, in}} - n_{\text{Pentene, out}}} \quad (7)$$

where $W_{\text{Pentene, in}}$ and $W_{\text{Pentene, out}}$ represent the mass flow rates of 1-pentene in the feed and the product, respectively; W_i and n_i are, respectively, the mass and mole flow rate of product i in the product stream; while $n_{\text{Pentene, in}}$ and $n_{\text{Pentene, out}}$ represent the mole flow rates of 1-pentene in the feed and the product, respectively. For more information on the evaluation of catalytic performance, please refer to ESI.

4. Conclusions

This work decodes the temporal blueprint of reaction pathways in high-temperature olefin cracking, moving the field beyond static catalyst descriptors. A micro fixed-bed reactor was employed to investigate the evolution of 1-pentene cracking pathways on H-ZSM-5 zeolite under typical TCO reaction conditions (650 °C). During the initial reaction stage, the CCR pathway leads to a relatively high ethylene content in the products. As the reaction time progresses, the contribution of the carbenium ion cracking pathway increases. This is because the C_5+ olefins formed from 1-pentene exhibit strong interactions with the H-ZSM-5 zeolite, facilitating their re-adsorption on the H-ZSM-5 surface for further cracking. As the reaction proceeds, enhanced thermal cracking of 1-pentene suppresses ethylene formation, while simultaneous aromatization of 1,3-butadiene yields additional aromatic products.

The interactions between C_2 – C_8 olefins produced during 1-pentene cracking and the H-ZSM-5 zeolite were examined through molecular dynamics (MD) simulations and computational analysis. Combining computational and experimental results, it is demonstrated that reducing the reaction time by increasing space velocity, shortening diffusion paths by decreasing zeolite crystallite size, and introducing mesopores to expand the average pore size of the zeolite are effective strategies to suppress side reactions and enhance the catalytic performance of the zeolite for 1-pentene cracking.

To enhance the predictive capability of the time-diffusion-selectivity framework, future research will focus on developing a comprehensive kinetic model that integrates the CCR, carbenium ion, and thermal cracking mechanisms with catalyst properties to guide the rational design of zeolite catalysts for olefin cracking.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/catal16030230/s1>, Figure S1: Comparison of XRD spectra of H-ZSM-5 zeolite and the refined structure results of the measured XRD spectra; Figure S2: FTIR spectrum of H-ZSM-5 zeolite; Figure S3: BJH pore size distribution curve and N_2 adsorption–desorption isotherm of H-ZSM-5 zeolite; Figure S4: SEM image of H-ZSM-5 zeolite; Figure S5: NH_3 -TPD curve, Py-FTIR spectrum and ^{27}Al MAS NMR spectra of H-ZSM-5 zeolite; Figure S6: Composition of 1-pentene cracking products at 650 °C; Figure S7: Adsorption density distribution of C_2 – C_8 olefins at 650 °C; Figure S8: Reaction rate constants for the cracking of C_5 to C_8 olefins at 650 °C; Table S1: Crystallographic Parameters of H-ZSM-5 Zeolite; Table S2: Assignment of characteristic peaks of H-ZSM-5 zeolite in FTIR; Table S3: Molar selectivity of 1-pentene cracking products at 650 °C; Table S4: Thiele modulus, reaction efficiency factor for the cracking reactions of C_5 – C_8 olefins on H-ZSM-5 zeolite, and optimal minimum particle size and pore diameter eliminating the influence of diffusion; Table S5: Sensitivity analysis of Thiele modulus (φ) and effectiveness factor (η) for C_5 – C_8 olefins; Text S1: Calculation method of DMR index;

Text S2: Calculation method of Thiele modulus (φ) and effective factor (η). References [6,43,59–67] are cited in the Supplementary Materials.

Author Contributions: Writing—original draft, S.H.; Investigation, S.H. and S.Z.; Methodology, S.H., L.L. and S.Z.; Data curation, S.H.; Conceptualization, S.H., Y.Z., L.L. and Y.X.; Validation, S.H.; Supervision, Y.Z. and Y.X.; Writing—review and editing, L.L.; Formal analysis, L.L.; Supervision, Y.X. All authors have read and agreed to the published version of the manuscript.

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